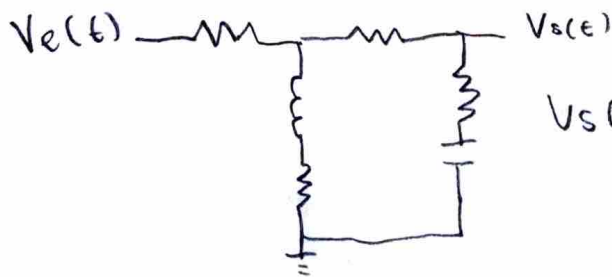


Ecuaciones Principales Análisis matemático



$$V_s(t) = R I_1(t) + L \frac{dI_1(t)}{dt} + R I_2(t) - \left[\frac{dI_2(t)}{dt} \right] = R I_2(t)$$

$$0 = R I_1(t) + L \frac{dI_1(t)}{dt} + R I_2(t) + R I_1(t) + \frac{1}{C} \int I_2(t) dt - R I_1(t) - \left[\frac{dI_2(t)}{dt} \right]$$

$$V_e(t) = R I_2(t) + \frac{1}{C} \int I_2(t) dt$$

$$V_s(s) = R I_1(s) + L s I_1(s) + R I_2(s) - \left[\frac{1}{Cs} I_2(s) - R I_1(s) \right]$$

$$0 = R I_2(s) + L s I_2(s) + R I_2(s) + R I_1(s) + \frac{1}{Cs} I_2(s) - R I_1(s) - L s I_1(s)$$

$$V_e(s) = I_1(s) [R + Ls + R] - I_2(s) [Ls + R]$$

$$0 = I_2(s) [R + Ls + R + R + \frac{1}{Cs}] - I_1(s) [R - Ls]$$

$$V_e(s) = I_1(s) [R + \frac{1}{Cs}] \quad V_s(s) = I_2(s) \left[\frac{R Cs + 1}{Cs} \right]$$

$$I_1(s) = \frac{I_2(s) [3R + Ls + \frac{1}{Cs}]}{R + Ls}$$

$$I_1(s) = I_2(s) [3R + Ls + \frac{1}{Cs}] \left[\frac{1}{R + Ls} \right]$$

$$I_1(s) = I_2(s) \left[\frac{3R}{R + Ls} + \frac{Ls}{R + Ls} + \frac{1}{Cs(R + Ls)} \right]$$

$$I_1(s) = I_2(s) \left[\frac{3R + Ls}{R + Ls} + \frac{1}{R Cs + L Cs} \right]$$

$$V_e(s) = I_2(s) \left[\frac{CLs^2 + 3RS + 1}{CLs^2 + Cs} \right] [2R + Ls] - I_2(s) [CLs^2 + 3RS + 1]$$

$$V_e(s) = I_2(s) \left[\frac{CLs^2 + 3RS + 1}{CLs^2 + Cs} \right] [(2R + Ls) - (Ls + R)]$$

$$V_e(s) = I_2(s) \left[\frac{2CLRs^2 + 3R^2 + 2R + CLs^3 + 3LCs^2 + (s) - (Ls + R)(CLs^2 + Cs)}{CLs^2 + Cs} \right]$$

$$V_e(s) = I_2(s) \left[\frac{3CLRs^2 + 3R^2 + CLs^3 + 2R + Ls - (CLs^3 + CLs^2 + CLs^2 + CLs)}{CLs^2 + Cs} \right]$$

$$V_e(s) = I_2(s) \left[\frac{3CLRs^2 + 3R^2 + 2R + Ls + CLs^3}{CLs^2 + Cs} \right]$$

Modelo de ecuaciones integrodiferenciales
 a partir de las ecuaciones principales del sistema, se despejan las variables dependientes

$$\frac{V_s(s)}{V_e(s)} = \frac{Cs^2 + 2}{Cs} \cdot \frac{(1,1)}{3CLs^2 + 3C\alpha^2 s^2 + 2\alpha + Ls + C\alpha^3} \cdot \frac{(1,1)}{CLs^2 + C\alpha s}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLs^2 + (C\alpha^2 + L)s + \alpha}{3CLs^2 + (5C\alpha^2 + L)s + 2\alpha}$$

$$L = 15 \mu H$$

$$R = 2.2 k\Omega$$

$$C = 270 nF$$

Error estacionario

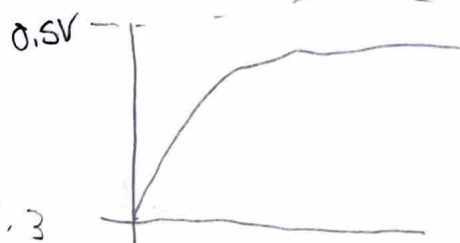
$$e(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \left(1 - \frac{CLs^2 + (C\alpha^2 + L)s + \alpha}{3CLs^2 + (5C\alpha^2 + L)s + 2\alpha} \right)$$

$$e(1) = \frac{\alpha}{2\alpha} \quad e(t) = 0.5V$$

estabilidad en lazo abierto $\lambda_1 = -1.8181$

$$-b \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$\lambda_2 = -11111109.3$$



El sistema se establece con una respuesta sobreamortiguada

Modelo de ecuaciones integrodiferenciales

$$V_s(t) = \alpha i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_1(t) [V_e(t) + \alpha i_2(t) + L \frac{d(i_1(t) - i_2(t))}{dt}]$$

$$i_2(t) = \left[L \frac{d(i_1(t) - i_2(t))}{dt} + \alpha i_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3}$$

$$V_i(t) = \alpha i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s = \frac{CLs^2 + (C\alpha^2 + L)s + \alpha}{3CLs^2 + (5C\alpha^2 + L)s + 2\alpha} = \frac{7.26 \times 10^{-6} s^2 + 1069.85 + 1100}{2.70 \times 10^{-5} s^2 + 5324 + 4400}$$